

Machine Learning for Brain MRI Analysis in Spinal and Bulbar Muscular Atrophy

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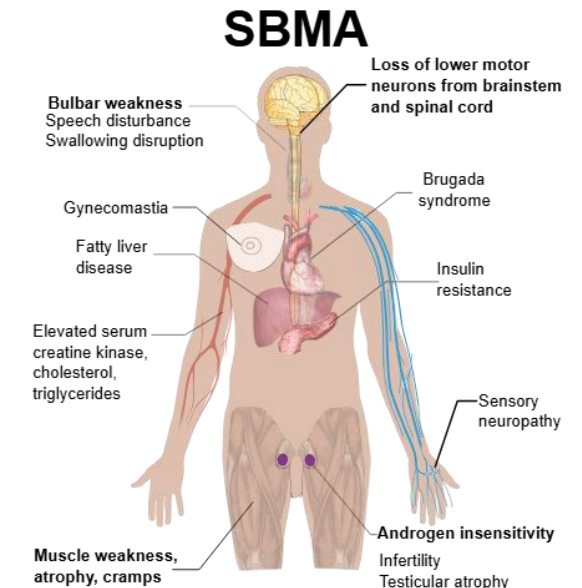
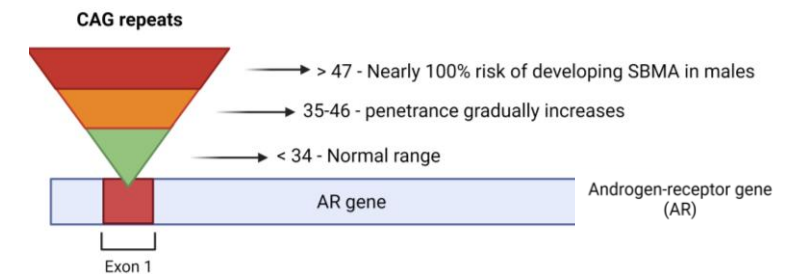
Supervisor
Francesco Rinaldi

Spinal and Bulbar Muscular Atrophy (SBMA)

A rare, X-linked disorder caused by CAG expansion in the androgen receptor gene. Leads to degeneration of motor neurons in the spinal cord and brainstem.

Symptoms

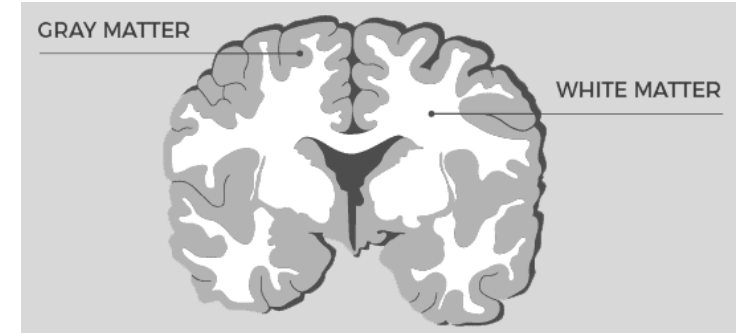
- Progressive muscle weakness and wasting, cramps, fasciculations and gait impairment ;
- Speech and swallowing difficulties ;
- Gynecomastia, erectile dysfunction, decreased libido, infertility, and testicular atrophy ;
- Metabolic disturbances.



Previous Research and Literature Gap

Neuroimaging studies:

- Well established white-matter atrophy in brainstem and spinal-cord.
- Limited and inconsistent findings on gray-matter involvement
 - No study has used machine learning techniques yet



Does SBMA also affect cortical gray matter?

1. Can we distinguish different communities in cortical regions organization between SBMA and healthy individuals? (community detection pipeline)
2. Can we correctly classify SBMA from healthy controls from cortical gray-matter images? (classification pipeline)
 - a. Which brain regions contribute the most to classification? (explainable AI methods)

Data

37

T1-weighted structural MRI
scans (3D voxel images)

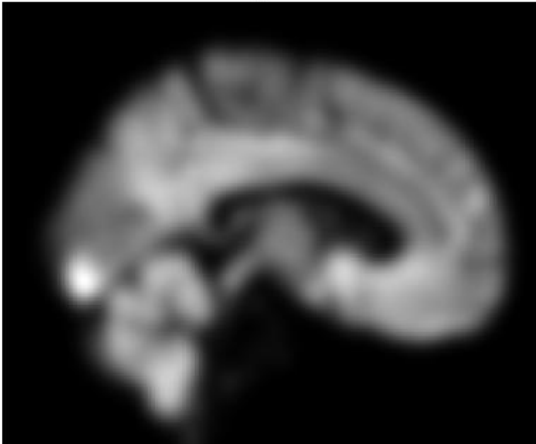
20

Participants with SBMA

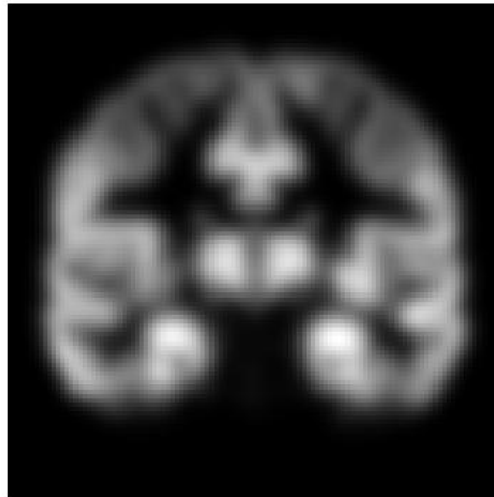
17

Healthy controls

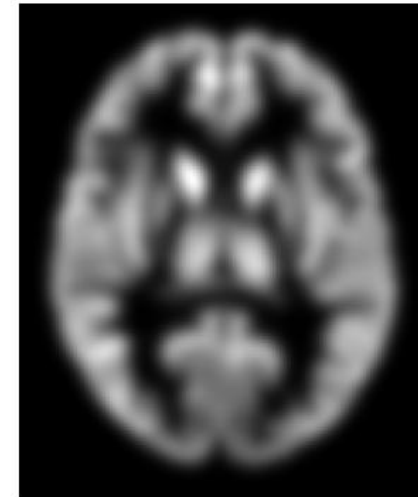
Sagittal



Coronal



Axial



Tissue segmentation, spatial normalization, modulation and smoothing with an 8-mm FWHM Gaussian kernel. (Done by neurologist).

Preprocessing

Images

Each preprocessed image contains $113 \times 137 \times 113$ voxels (= 1,749,353). Voxels have a gray-matter scale value ranging from 0 to 1.

Parcellation

Schaefer parcellation: 1,000 parcels, 500 per hemisphere

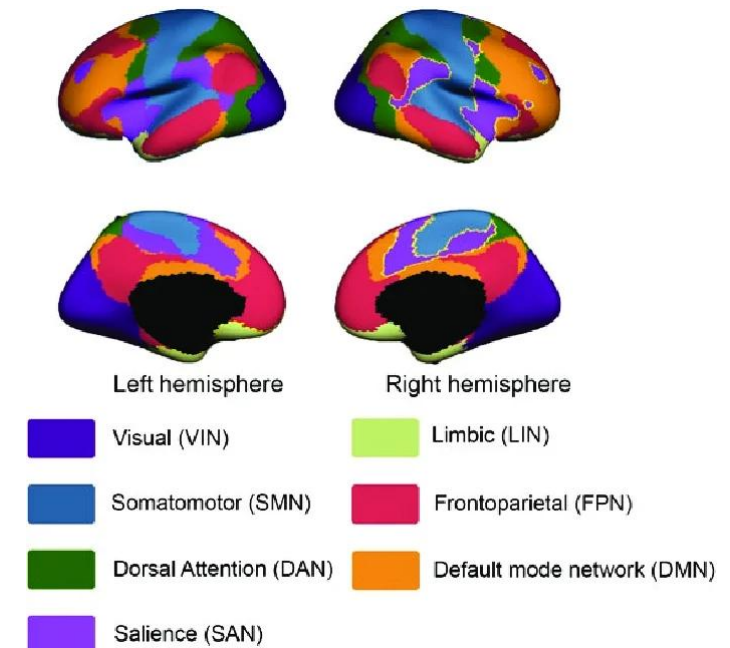
Each parcel belongs to one of seven canonical functional networks from Yeo et al.

- To each parcel we assigned the mean gray-matter value of the voxels belonging to the parcel

Mean gray-matter(GM) in parcel j for participant s:

$$x_{sj} = \frac{1}{|V_j|} \sum_{v \in V_j} GM_s(v)$$

Resulting input matrix (37x1000)



Community Detection Pipeline

subject graphs

Each participant has an undirected, weighted graph.

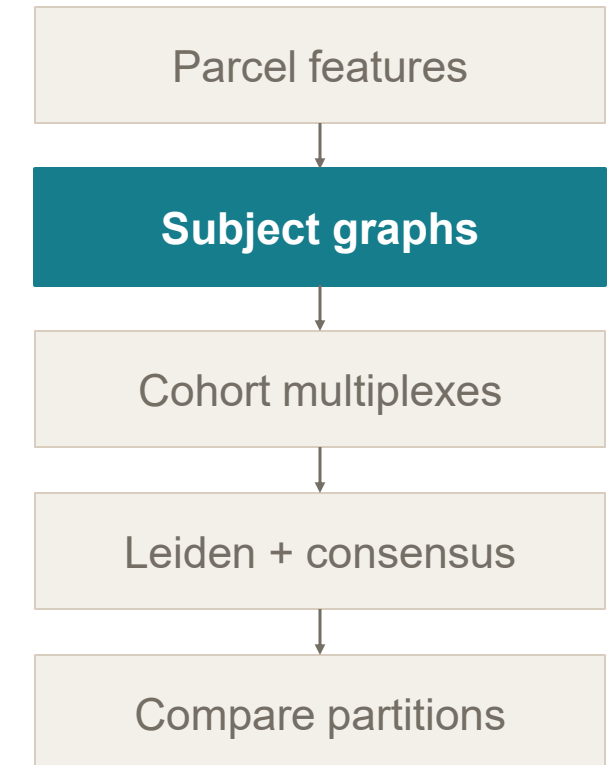
- Nodes: per parcel
- Edges: Connect centroids whose Euclidean distance is at most 15 spatial units.
- Edge weights : RBF similarity between connected nodes

$$w_{ij} = \exp \left[-150(x_i - x_j)^2 \right]$$

Mean degree 26.97

Median 27, range 7–58 connections

COMMUNITY DETECTION PIPELINE

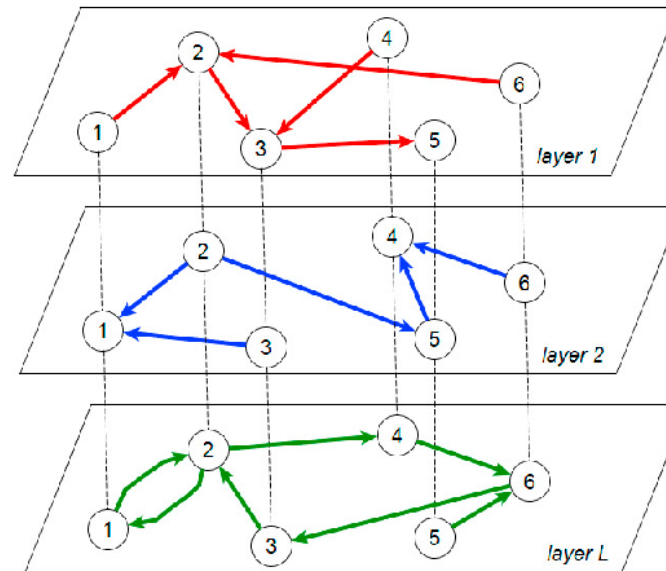


Community Detection Pipeline

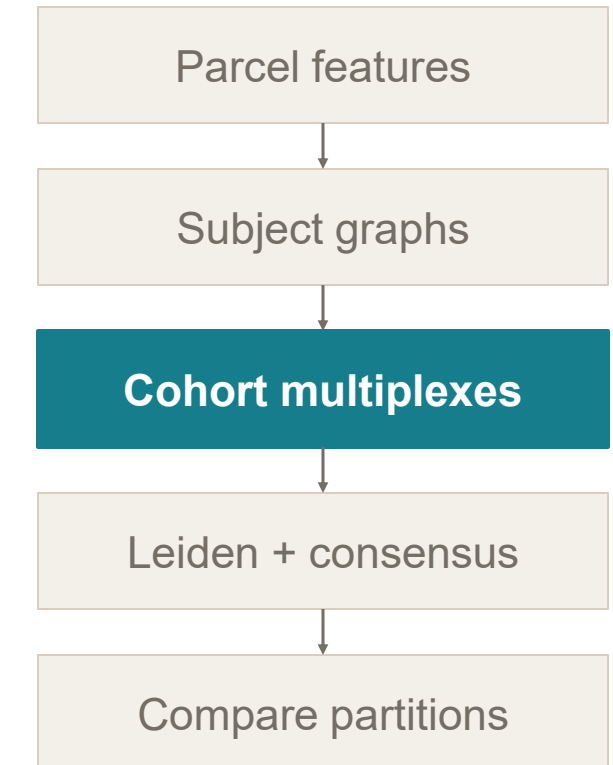
Cohort multiplex

- 1 multiplex for SBMA
- 1 multiplex for healthy controls

The same 1,000 parcel identities appear in every layer, with participant-specific GM similarity weights.



COMMUNITY DETECTION PIPELINE



Multiplex	Layers
Healthy controls	17 subject graphs
SBMA	20 subject graphs

Community Detection Pipeline

Leiden and consensus clustering

Leiden is a greedy community-detection algorithm based on modularity optimization.

Step 1. Local Moving of Nodes

Step 2: Refinement of the Partition

Step 3: Aggregation of the Network.

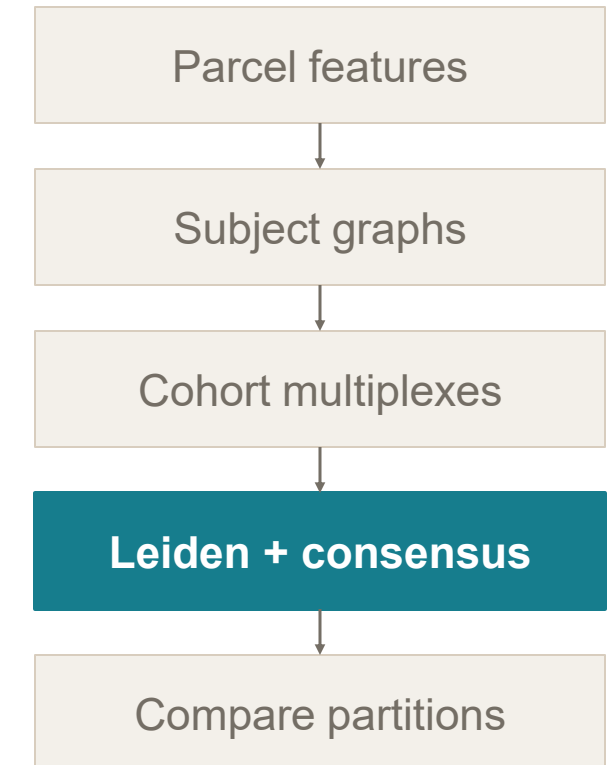
$$Q_k(\gamma) = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \gamma \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

$$Q = \sum_k \lambda_k Q_k$$

$$C_{ij} = \frac{1}{300} \sum_r I(c_i^{(r)} = c_j^{(r)})$$

1. We run the algorithm 30 times for 10 different resolution values ($\gamma = 0.1, 0.2, \dots, 1.0$) and created a consensus matrix:
2. Retain consensus connections with $C_{ij} \geq 0.7$
3. Build a graph from the consensus matrix
4. Re-run Leiden until two consecutive partitions are identical

COMMUNITY DETECTION PIPELINE



Community Detection Pipeline

Compare partitions

Overall agreement

Adjusted Rand Index (ARI)

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{1 - \mathbb{E}[\text{RI}]}$$

Normalized Mutual Information (NMI)

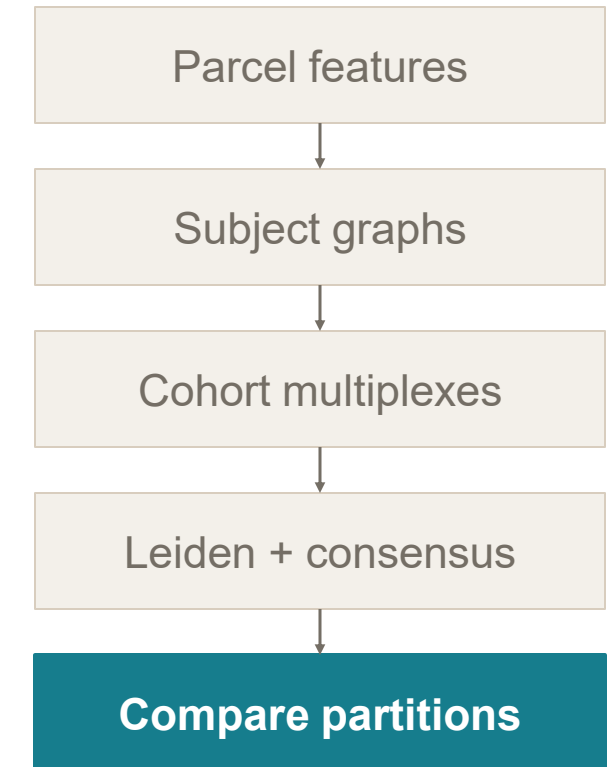
$$\text{NMI}(U, V) = \frac{2I(U; V)}{H(U) + H(V)}$$

Clusters overlap

Jaccard similarity

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

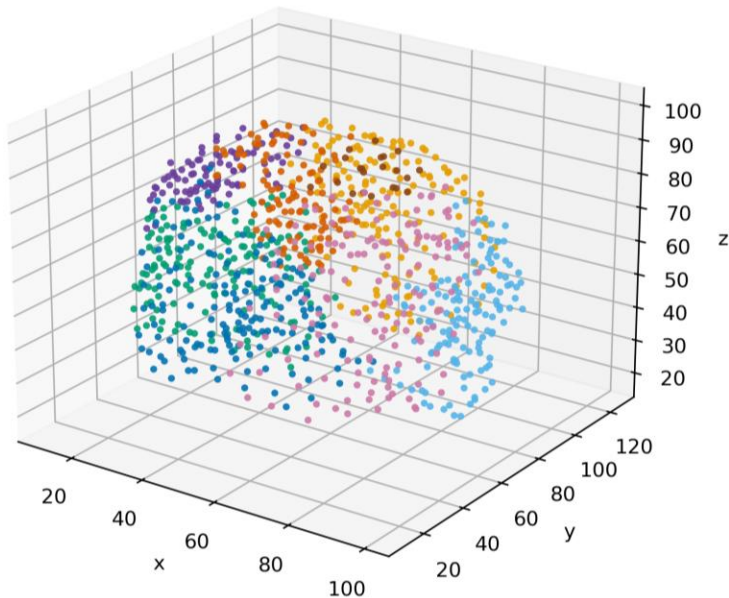
COMMUNITY DETECTION PIPELINE



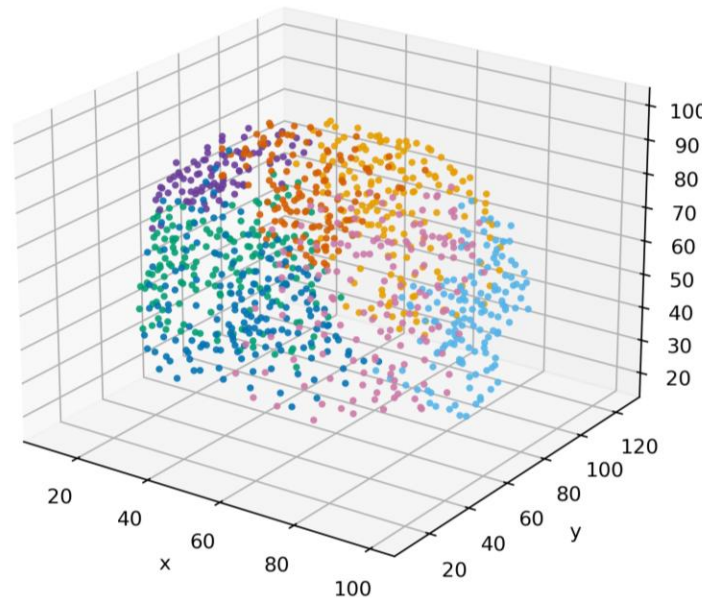
Results:

Overall community organization is similar

Healthy controls



SBMA participants



8 vs 7

HC vs SBMA
communities

0.962

Adjusted Rand Index

0.960

Normalized Mutual
Information

Colours align communities by their highest Jaccard match.

Results:

Most communities have a close counterpart

HC / SBMA	0	1	2	3	4	5	6
0	.98	.00	.00	.00	.00	.00	.00
1	.00	.00	.98	.00	.00	.00	.00
2	.00	.95	.00	.00	.00	.00	.00
3	.00	.00	.00	1.00	.00	.00	.00
4	.00	.00	.00	.00	.99	.00	.00
5	.01	.00	.00	.00	.00	.84	.00
6	.00	.00	.01	.00	.00	.00	.96
7	.00	.03	.00	.00	.00	.12	.00

Jaccard similarity

Six HC communities have a match of at least 0.95.

HC community 5 has a match of 0.84.

HC community 7 is redistributed across two SBMA communities.

Labels are community IDs. Similarity ranges from 0 (no overlap) to 1 (identical parcel sets).

Results:

The main difference involves HC community 7

HC community 7	Parcels
Left somatosensory–motor network	19
Left dorsal attention: frontal eye field subdivision	2
Left dorsal attention: posterior subdivision	1
Total	22

16

Parcels assigned to SBMA community 5

6

Parcels assigned to SBMA community 1

Seven additional parcels change assignment outside HC community 7.

These exploratory differences do not establish disease-specific reorganization on their own.

Classification pipeline

input representations

- **Raw regional gray-matter:**

- x_{sj} is the average GM value within parcel j for participant s .

$$x_{sj} = \frac{1}{|V_j|} \sum_{v \in V_j} GM_s(v)$$

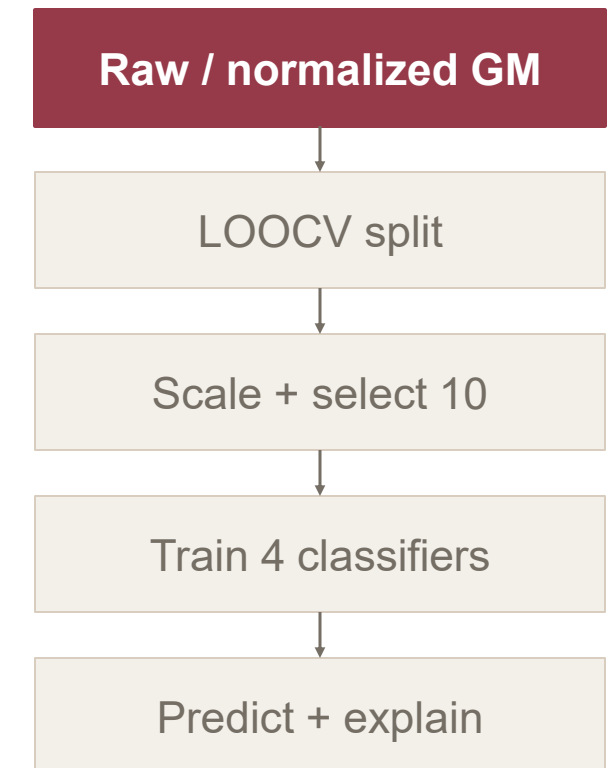
- **Normalized regional gray-matter**

- x_{sj} is normalized by the total gray-matter of the subject T_s .
- Normalized regional gray-matter expresses each parcel mean relative to total image gray-matter.

$$T_s = \sum_{v \in \text{image}} GM_s(v), \quad \tilde{x}_{sj} = \frac{x_{sj}}{T_s}$$

Both representations contain 1,000 regional features.

CLASSIFICATION PIPELINE



Classification pipeline

Cross-validation and feature selection

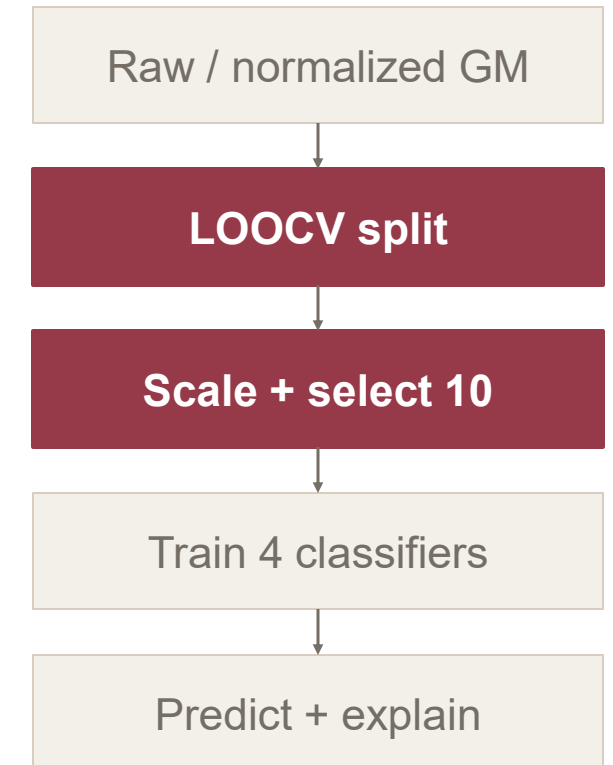
37 leave-one-out folds:

Training set: 36 participants

Test set : 1 participant

1. Fit a standard scaler to the training set
2. Rank features by mutual information
 - a. Keep the 10 highest scoring feature
3. Apply the fitted scaler and the same feature subset to the held-out participant

CLASSIFICATION PIPELINE



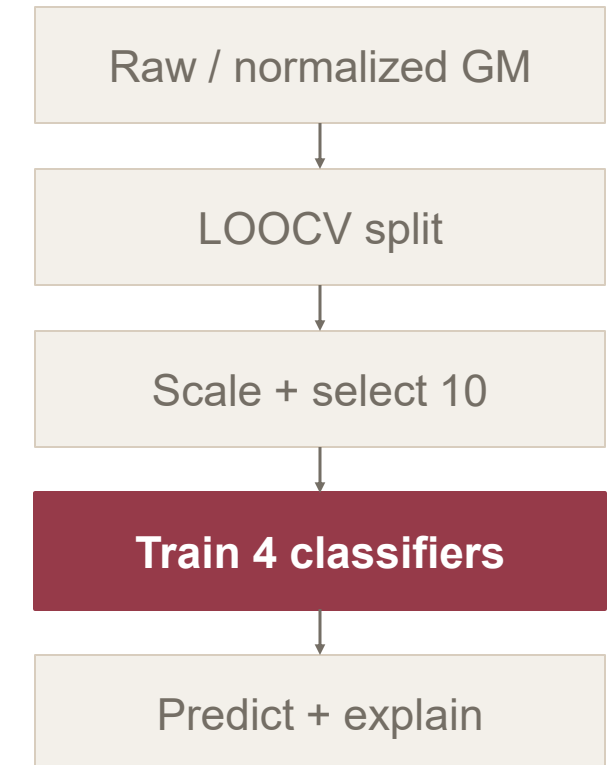
Classification pipeline

model fitting

Classifier	Configuration
Support Vector Machine	SVC with a linear kernel
Logistic Regression	L1 penalty, liblinear solver
Random Forest	random_state = 4
XGBoost	XGBClassifier defaults

Fit all four classifiers on both input representations within the same 37-fold evaluation.

CLASSIFICATION PIPELINE



Classification pipeline

Explainable AI

Compute performance metrics

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

F1

$$\text{F1-score} = \frac{2TP}{2TP + FP + FN}$$

Sensitivity

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

Specificity

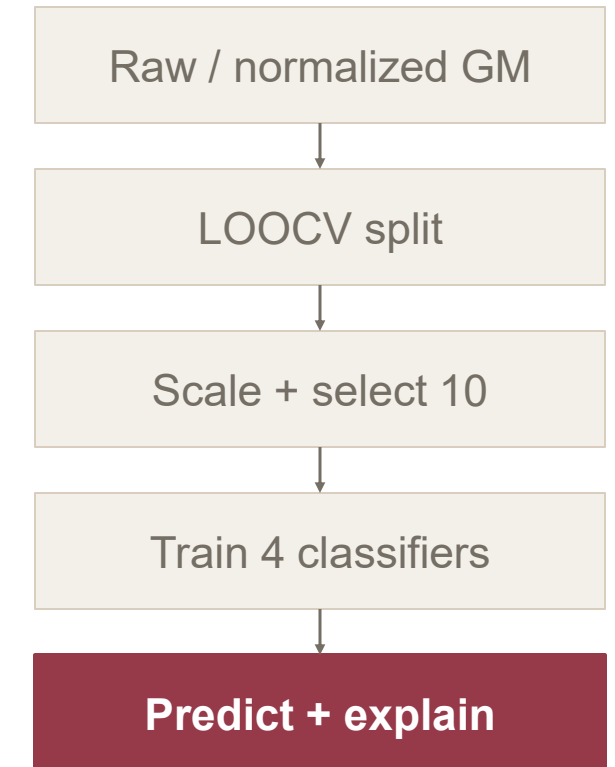
$$\text{Specificity} = \frac{TN}{TN + FP}$$

TP = true positive,
TN = true negative,
FP = false positive,
FN = false negative.

Compute SHAP for each held-out participant

Positive SHAP values favour SBMA. Negative SHAP values favour HC.
Unselected parcels receive zero.

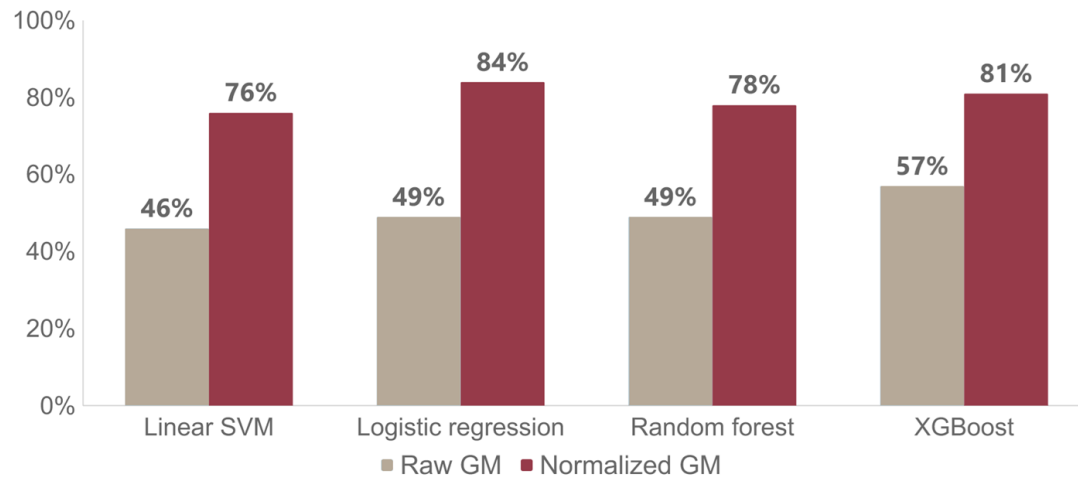
CLASSIFICATION PIPELINE



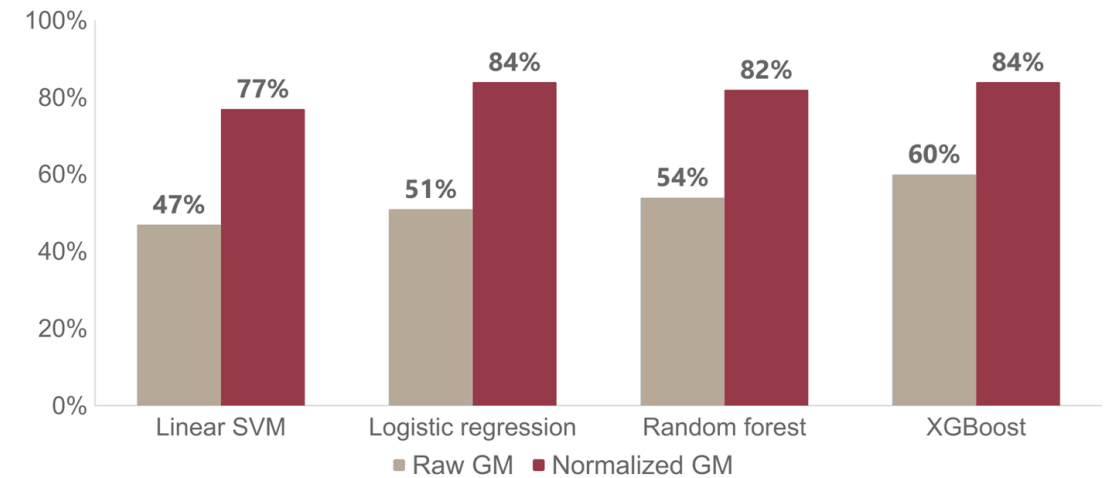
Results:

classification accuracy

Classification accuracy



F1-score for SBMA



84%

Highest accuracy and F1 with
L1 logistic regression

Results:

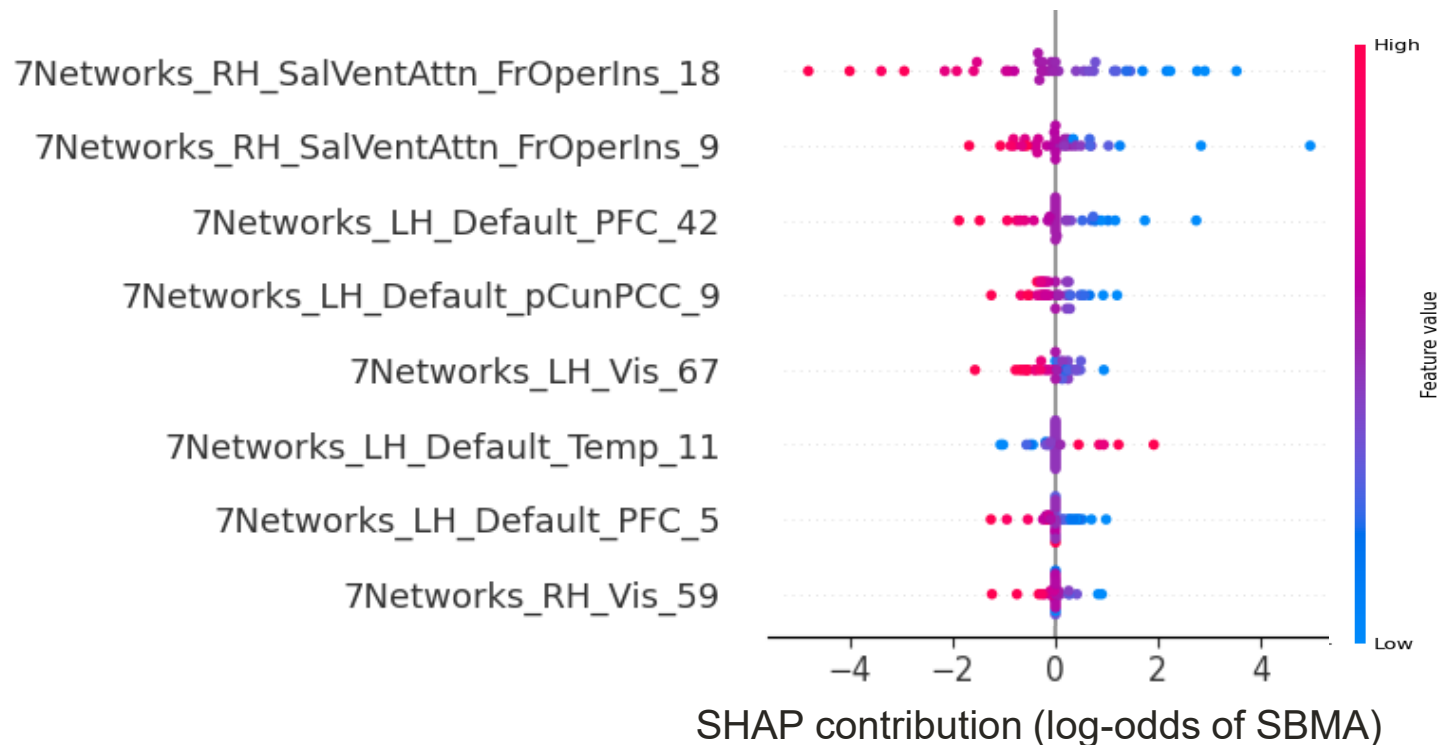
classification accuracy

Normalized GM	Accuracy	F1 (SBMA)	Sensitivity	Specificity
Linear SVM	0.76	0.77	75.0%	76.5%
Logistic Regression	0.84	0.84	80.0%	88.2%
Random Forest	0.78	0.82	90.0%	64.7%
XGBoost	0.81	0.84	90.0%	70.6%

Results:

Strongest regional contributors

Top eight contributors



Logistic Regression

Lower normalized GM in right frontal operculum–insula and left prefrontal parcels generally favours SBMA.

Left temporal parcel 11 shows the opposite pattern.

One point per held-out participant.

All four classifiers rank the first three parcels as the most influential.

These are predictive associations, no real atrophy measures.

Summary of results

	Community detection	Classification
Question	Can we distinguish different communities in cortical regions organization between SBMA and healthy individuals?	Can we correctly classify SBMA from healthy controls from cortical gray-matter images? Which regions contribute the most to classification?
Main finding	Broad agreement between cohorts, with localized somatomotor (and dorsal attention) differences	We can distinguish groups with accuracy up to 84%(with normalized gray-matter). Discriminative regions are frontal operculum–insula in ventral attention and default prefrontal cortex .

Results interpretation:

Methodological explanation

	Community detection	Classification
Unit of analysis	Relationships among parcels	Individual participants
What is measured	Similarity (within a group)	Differences (between groups)

A parcel can help classify a participant while retaining its community membership.

Results interpretation:

Biological explanation

SBMA may affect different parcels within the somatosensory–motor network across individual patients.

- 91 parcels in LH and 103 parcels in RH hemispheres.
- Which parcel present more atrophy might be patient specific.

Limitations

Small sample size and many candidate features

37 participants limit the precision and generalizability of performance estimates.

Univariate feature selection

Mutual information ranks parcels separately and may miss interactions.

Graph construction

Distance, RBF width and consensus settings influence communities.

Future research

Next step	What it would clarify
Larger, independent samples	Reproducibility of classification
Fit a linear model to predict parcels expected values and use residual regional gray-matter (can account for confounding variables as age)	Disease-related variation beyond expected (age related) differences
Functional MRI based graph parameters	Less influenced by hyperparameter choice. We could look at patterns of brain activation.

Conclusions

Cortical organization

The cohorts share similar communities, with localized differences centred on the left somatosensory–motor system.

Discrimination

Normalized regional gray-matter supports internal classification accuracy up to 84%.

Regional contribution

Frontal operculum–insula (parcels 18,9) and prefrontal (parcel 42) emerge as candidate regions for further study.

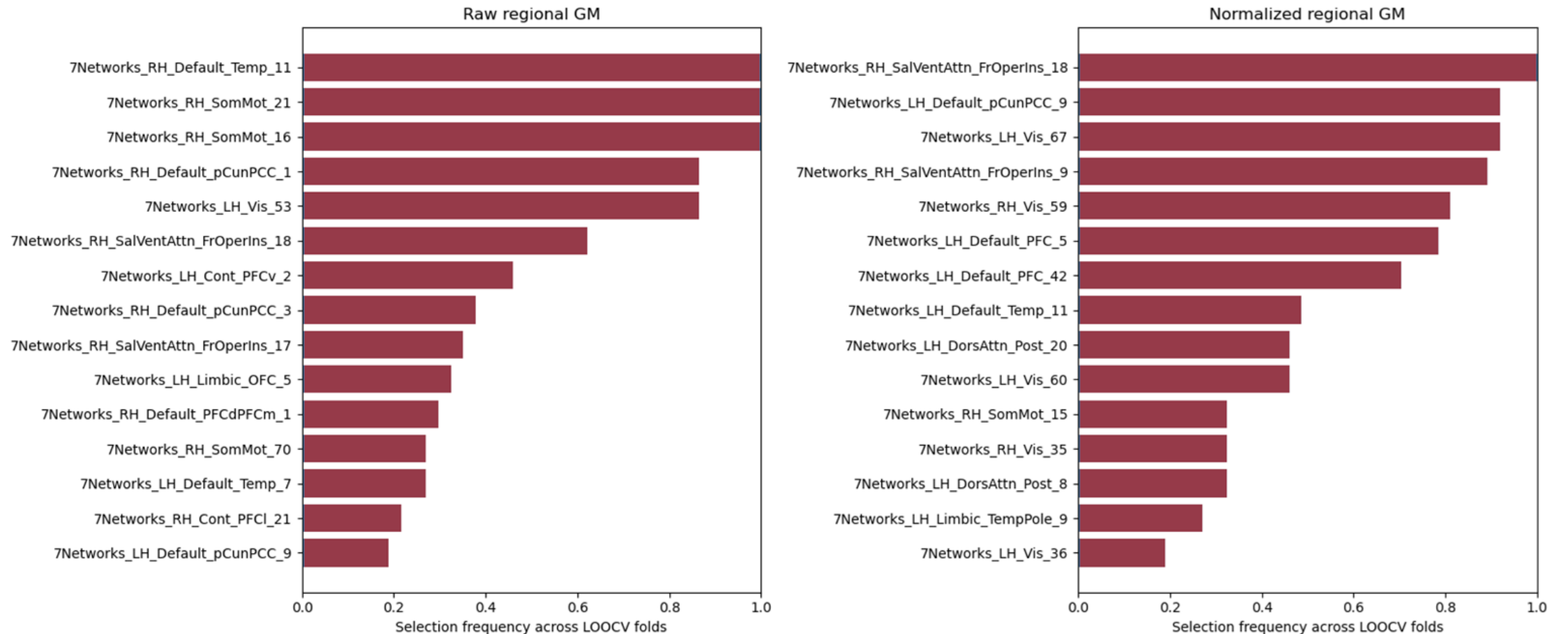
Thanks for the attention

Giorgia Bonin

Results:

Selection frequency

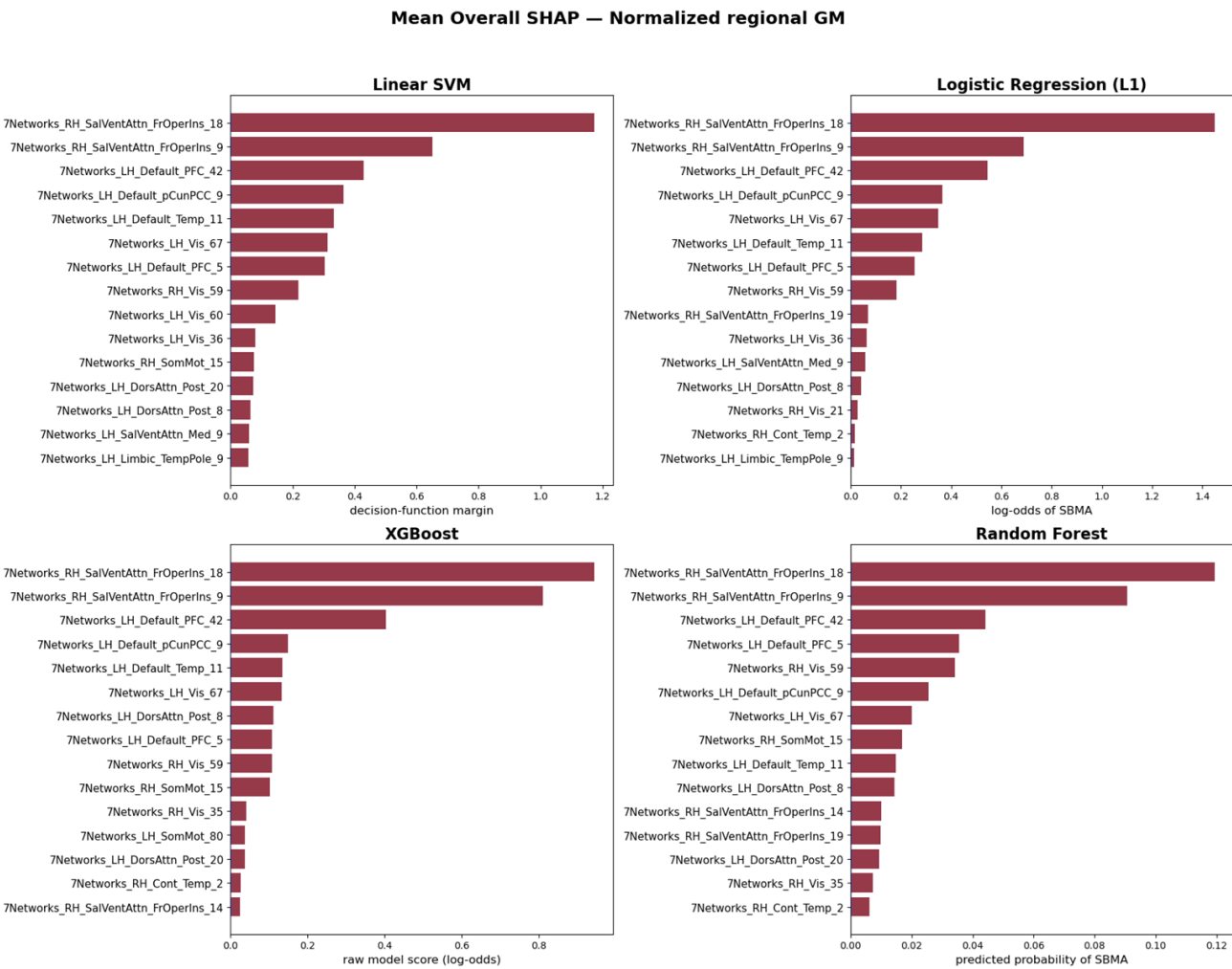
Top 15 selected parcels for each GM representation (K=10)



Results:

Mean Overall SHAP

From here on only results related to the normalized gray-matter input will be analysed.

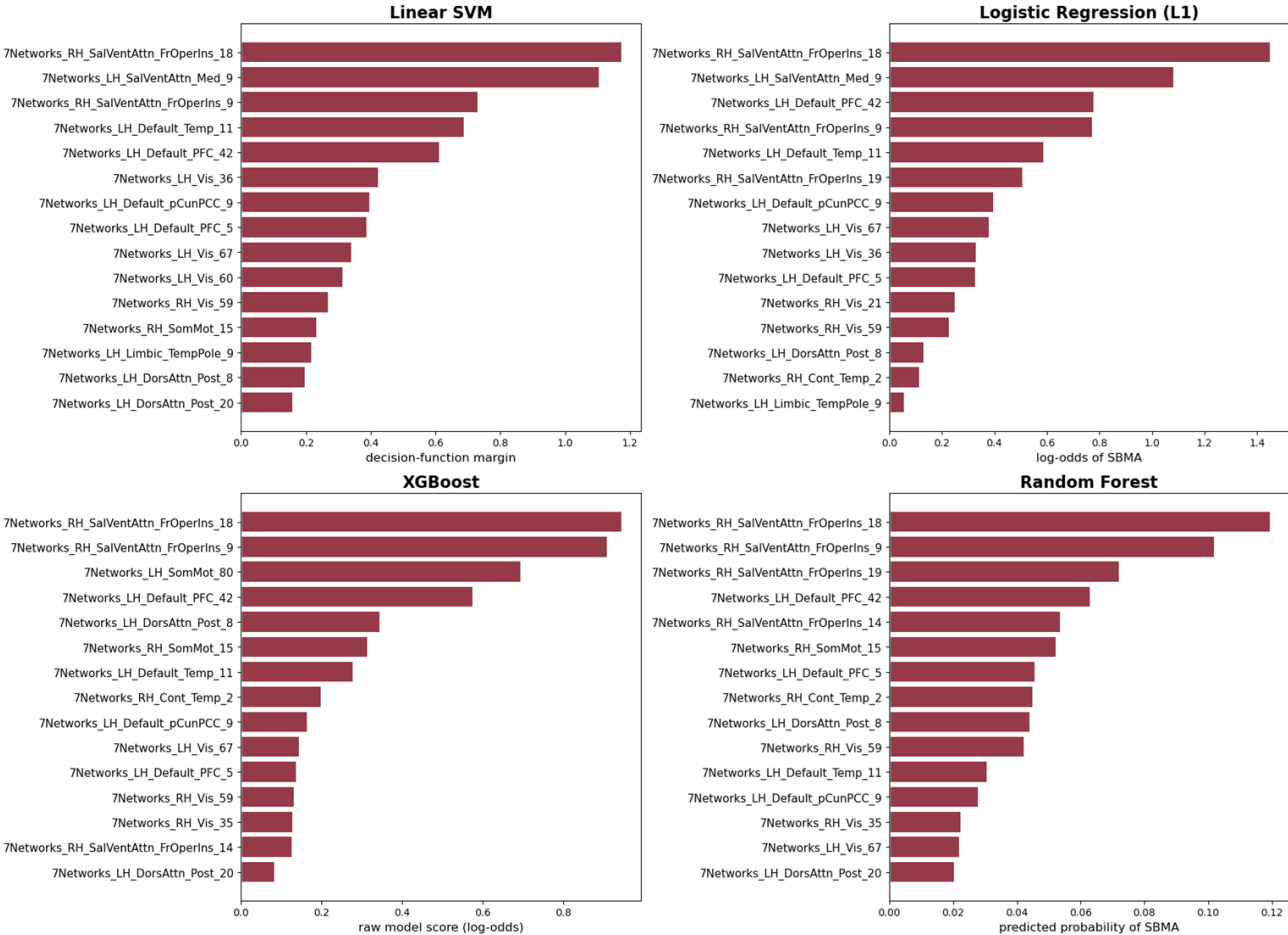


Results:

Mean Absolute SHAP when Selected

From here on only results related to the normalized gray-matter input will be analysed.

Mean Absolute SHAP When Selected — Normalized regional GM



Results:

Mean Signed SHAP

From here on only results related to the normalized gray-matter input will be analysed.

